Solution Requirements

Date	29 September 2026
Team ID	SWTID-2026-8642
Project Name	Import Data Using Transform Maps (Spreadsheet)
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	**Authentication**	Users must securely log into the system using Role-Based Access Control (RBAC) and Multi-Factor Authentication (MFA) before accessing the data import module.	**High**
2	**Authorization levels**	**Admin:** Configures Transform Maps, scripts, and target table schema.	**High**

		Data Analyst: Uploads spreadsheets and runs transform processes. **Viewer:** Views import status, logs, and report dashboards.	
3	**External interfaces**	The system must integrate with Microsoft Excel / CSV file parsers, target application databases (e.g., ServiceNow/SQL), and cloud storage services.	**Medium**
4	**Transactions processing**	Parses source spreadsheet data, matches columns to target fields via Transform Maps, handles coalesce (unique key) matching, and executes data inserts or updates.	**High**
5	**Reporting**	Generates real-time import summary reports, log dashboards showing total rows processed, skipped records, and detailed error logs for failed rows.	**Medium**
6	**Business rules, etc.**	• Automatically ignores duplicate rows based on Coalesce key parameters. • Rejects rows with missing mandatory fields. • Runs custom pre/post transform scripts for data cleaning.	**High**

7	**Compliance to laws or regulations**	Must comply with **GDPR** and **SOC 2** data privacy standards for anonymizing and masking Sensitive Personal Identifiable Information (PII) during upload.	**Medium**
8	**Other**	Supports one-click re-transformation for failed or rejected records without requiring a full file re-upload.	**Low**

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	**Performance & Speed**	File upload, parsing, and transform mapping execution speed for standard spreadsheets.	Processes a file of 10,000 records in < 30 seconds; page response time < 2 seconds.
2	**Scalability**	System capability to process large-volume dataset files without crashing or timing out.	Supports multi-threading and chunking for files up to **100 MB** or **1,000,000 rows**.
3	**Security & Data Privacy**	Encryption standards for uploaded spreadsheet files both at rest and in transit.	**AES-256** encryption for stored files and **TLS 1.3** protocol for file transfers.
4	**Reliability & Availability**	System operational uptime and automated	**99.9% uptime** per month; zero database corruption

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
		error handling during batch import failures.	on system failures (Atomic transactions).
5	**Usability & Accessibility**	User-friendly drag-and-drop file upload UI and intuitive auto-mapping screen setup.	**WCAG 2.1 AA** compliance; completion of mapping setup within < 3 clicks for standard templates.
6	**Other (Maintainability)**	Code modularity and log retention for auditing import histories over time.	Retains detailed import set execution logs for at least **90 days**; automated daily log purges.